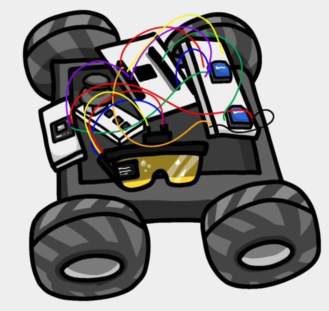
HEATSeeker

Ruey Day, Yi Keen Lim, Jungeun Seo, Sophie Yang
{rueyday, yklim, sjungeun, sophy}@umich.edu

EECS 373

Overview & Core Functionality



Our search-and-rescue embedded system provides a thermal view and robot status directly in smart glasses, while the robot can be driven with a joystick or automatically track the hottest object in view.

- Robot drives in hazardous, low-visibility spaces
- Thermal camera finds heat sources (e.g., person, hotspot)
- Glasses show a live heatmap + battery + mode so the user never has to look to a separate screen
- Mixed control: user can joystick steer or let the robot auto-track

Problem Statement: Safer Navigation

Problems

- Smoke, darkness, debris and structural instability make it dangerous for human to enter the buildings.
- Existing solution often use separate monitors and completely manuel driving, forcing the operators to constantly look away.

What is required

- Continuous thermal awareness
- Simple, intuitive control
- Option to automate small tasks (like tracking the hottest object)

User Joystick

SActual Picture of the Components

SSystem diagram

Xbee Msg typereply on discord

Xbee (2 bytes)

Rescue Robot

SActual Picture of the Components

SSystem diagram

- Differential-drive platform
- Proportional controller that turns toward the hottest pixel cluster
- Wireless telemetry back to the glasses

Xbee Msg type Ignore if we don't have time

Xbee (34 bytes)

HUD Glasses

SActual Picture of the Components

SSystem diagram

- Displays:
  - Real-time thermal heatmap
  - Battery percentage (onboard Li-ion pack)
  - Current control mode (Joystick / IR / Mixed)
- Lightweight XBee link to receive data from the robot
- Simple physical buttons on the frames for mode/scale changes (if desired)

Tracking Control

- Feedback proportional control system that correct for errors based on the hottest pixel from thermal camera.
- Correct x error by going straight and y error by rotation.

More Details and Demos @ https://rueyday.github.io/HEATSeeker/